MATH/CSC 4610/6610 Numerical Analysis I – Course Syllabus

Instructor
Xiaojing Ye (xye@gsu.edu)

Office Hours Time and Location
MW 1:30-2:30 PM or by appointment. 25 Park PL 1436A.

Lecture Time
MW 12:00–1:15 PM.

Description
Introductory concepts and calculus review, foundation of Matlab programming, the sources and propagation of errors, root finding for nonlinear equations, interpolation and approximation theory, numerical integration and differentiation.

Prerequisites
Grade of C or higher in Math 2215 Multivariate Calculus or equivalent and the ability to program in a high-level language.

Textbooks
Numerical Analysis, Burden and Faires, 9th edition.

Administrative Drop Policy
Students who did not take the required prerequisite or do not attend the class regularly during the first two weeks will be administratively dropped.

Procedures
Class meets twice a week. Taking good notes during the class is of paramount importance. Homework will be assigned in each class. After the class read the book, read your notes and do as many of the homework problems as you can prior to the next class. Try to get the remaining problems explained in the next class or during the office hours. **You are responsible for all material covered in class, whether or not you attended this class**.

Team Competitions
There will be 5 team competitions (2 points each) during the semester. The purpose of these competitions is to take attendance and to keep you up-to-date in the course. **Make-up competitions will not be given**, except when special conditions exist.

Examinations
There will be 2 hourly exams (20 points each) and the final exam (40 points). All hourly exams will be taken during the regular class time and in the regular classroom. **Books and notes will not be allowed on all tests**. Missed exams will receive a score of 0. **Any conflicts must be worked out ahead of time. There will be no make-up exams after the test days except in an extreme verifiable emergency (e.g., a doctor's note is necessary)**. The tests and the final for graduate students (Math & CSC 6610) will contain additional problem(s).

Computer Projects
There will be 2 computer projects (5 points each) during the semester. Both projects will be given in MATLAB, which is a simple and powerful mathematical package. Deadlines for the computer projects will be announced in class. **Late projects will not be graded.**

Grading
There will be a total of 100 points possible for this course: Competitions (10pts). Computer Projects (10pts). Midterm Exam 1 (20pts). Midterm Exam 2 (20pts). Final (40pts).
A+ (97-100); A (93-96); A- (90-92); B+ (87-89); B (83-86); B- (80-82); C+ (77-79); C (70-76); D (60-69); F (<60).

Withdrawal
If you withdraw from this class on or before Midpoint day, you will receive a PW. If you hardship withdraw after this date you will receive a WF unless your average is 70% or higher (http://deanofstudents.gsu.edu/student-assistance/emergency-withdrawal/).

Cheating/Plagiarism
Plagiarism and cheating are serious offenses and may be punished by failure on the exam/project. Repeated cheating will result in a grade of F for the course. See the University's policy on Academic Honesty: (http://deanofstudents.gsu.edu/student-conductpolicy-on-academic-honesty/).

Conduct Policy
Any type of inappropriate conduct may result in your being administratively dropped from the course. See the University's Policy on Disruptive Behavior in the General Catalog, page 24 (http://codeofconduct.gsu.edu/files/2017/07/2017_18_6_29_17_Code.pdf).

Inclement Weather Policy
If the University is closed due to inclement weather, any exam that may have been scheduled for that date will be administered on the next available class date. If an assignment is due that day (or if you are scheduled for an exam), it will be due the next class.

Studying
You must work on this course every week. The pace is hectic and allowing yourself to fall behind will end in disaster.

This course syllabus provides a general plan for the course; deviations may be necessary.

MATH/CSC 4620/6620 Numerical Analysis II – Course Syllabus

Instructor
Xiaojing Ye (xye@gsu.edu)

Office Hours Time and Location
MW 1:30-2:30 PM or by appointment. 25 Park PL 1436A.

Lecture Time
MW 12:00–1:15 PM.

Description
Matlab programming; Numerical techniques for ordinary differential equations: implicit methods and predictor-corrector schemes, Runge-Kutta methods, the Adams Families; Gaussian Elimination for linear systems, the LU factorization, stability, SPD matrices and Cholesky Decomposition, iterative methods for linear and nonlinear systems; Finite difference methods for partial differential equations.

Prerequisites
Grade of C or higher in Math 2215 Multivariate Calculus or equivalent and the ability to program in a high-level language.

Textbooks
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Administrative Drop Policy
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Procedures
Class meets twice a week. Taking good notes during the class is of paramount importance. Homework will be assigned in each class. After the class read the book, read your notes and do as many of the homework problems as you can prior to the next class. Try to get the remaining problems explained in the next class or during the office hours. **You are responsible for all material covered in class, whether or not you attended this class**.

Team Competitions
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Examinations
There will be 2 hourly exams (20 points each) and the final exam (40 points). All hourly exams will be taken during the regular class time and in the regular classroom. **Books and notes will not be allowed on all tests**. Missed exams will receive a score of 0. **Any conflicts must be worked out ahead of time. There will be no make-up exams after the test days except in an extreme verifiable emergency (e.g., a doctor's note is necessary)**. The tests and the final for graduate students (Math & CSC 6610) will contain additional problem(s).

Computer Projects

There will be 2 computer projects (5 points each) during the semester. Both projects will be given in MATLAB, which is a simple and powerful mathematical package. Deadlines for the computer projects will be announced in class. **Late projects will not be graded.**

Grading
There will be a total of 100 points possible for this course: Competitions (10pts). Computer Projects (10pts). Midterm Exam 1 (20pts). Midterm Exam 2 (20pts). Final (40pts).
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